

Multiview Consistency Learning with Synthetic Camera Views for Hand-in-the-Wild Gesture Recognition

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Abstract—Hand gesture recognition is an important problem in human–computer interaction, virtual reality, augmented reality, robotics, and multimedia systems. Despite strong performance in previous studies, recognition in uncontrolled environments remains challenging due to viewpoint variation, self-occlusion, articulation ambiguity, and cross-dataset domain shift. To address these issues, we propose a 3D skeleton-based synthetic multi-view framework. A Blender-based animation pipeline reconstructs hand poses from 3D skeleton annotations, while multiple virtual cameras generate camera-space skeleton views from different viewpoints. These synthetic views are used only during training, preserving single-view inference. The original skeleton and its generated views are processed by a shared GRU encoder, and an original-anchored consistency loss aligns their representations. Experiments are conducted under a challenging 7-class cross-dataset setting, using *CanonicalSet* for training and *HandinWildSet* for external evaluation. The proposed 8-camera original-anchored consistency framework achieves the highest accuracy of 75.85% on *HandinWildSet*, outperforming the GRU baseline (62.96%) and rotation-based augmentation (70.22%). The corresponding 8-camera model without consistency achieves 68.44%, confirming the additional benefit of original-anchored consistency. Experimental results demonstrate that combining synthetic camera-space views with consistency learning improves viewpoint robustness and cross-dataset generalization.

Index Terms—hand gesture recognition, 3D skeleton, multiview learning, Blender, GRU

I. INTRODUCTION

Hand gesture recognition provides a natural interface for human–computer interaction, virtual reality, robotics, and multimedia systems [1], [2]. In this work, we focus on improving in-the-wild static hand gesture recognition, which remains challenging due to viewpoint variation, self-occlusion, and finger articulation ambiguity. Under certain viewpoints, different gestures may appear visually similar in uncontrolled conditions [4], [5]. As illustrated in Fig. 1, a *thumbdown* gesture may resemble *rock*, while *scissors* may appear similar to thumb-related gestures when finger visibility is limited.

To investigate this issue, we construct two datasets for cross-dataset hand gesture recognition. The first dataset, denoted as *CanonicalSet*, contains relatively clean hand-pose samples with standardized gesture appearances, whereas *HandinWildSet* includes samples captured under more practical conditions with varying viewpoints, hand orientations, and less controlled acquisition settings. Such differences create a substantial distribution gap between the two datasets. In particular, skeleton-

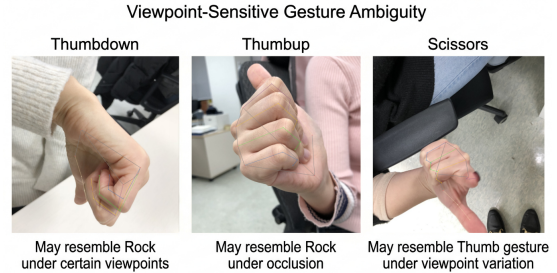


Fig. 1: Examples of viewpoint-sensitive hand gesture ambiguity in hand-in-the-wild conditions.

based methods remain sensitive to viewpoint changes because the same gesture can produce markedly different coordinate distributions when observed from different camera perspectives. Consequently, models trained on limited viewpoints may learn viewpoint-specific patterns rather than robust viewpoint-invariant gesture representations.

To address this problem, we propose a multi-view hand representation learning framework that combines synthetic camera-space skeleton generation with original-anchored consistency learning. A Blender-based hand animation pipeline first reconstructs hand articulation from 3D skeleton annotations, after which multiple virtual cameras generate camera-space skeleton observations from different viewpoints. These synthetic views are used only during training as viewpoint supervision, while inference requires a single observed skeleton. The original skeleton and its synthetic views are processed by a shared GRU encoder, and an original-anchored consistency loss aligns their embeddings to encourage viewpoint-invariant representation learning. The main contribution of this work is the integration of virtual-camera-based camera-space skeleton generation with original-anchored consistency learning for training-time multiview supervision while preserving single-view inference. The contributions of this paper are summarized as follows:

- We propose an original-anchored multiview consistency learning framework that preserves single-view inference while leveraging synthetic camera-space views generated by a Blender-based hand animation and virtual-camera pipeline during training.

- We systematically evaluate multiple recognition strategies under cross-dataset settings using *CanonicalSet* and *HandinWildSet*. The experimental results demonstrate that synthetic multiview data substantially improves viewpoint robustness and cross-dataset performance in hand-in-the-wild gesture evaluations.

II. RELATED WORK

A. Skeleton-based hand gesture recognition

Skeleton-based hand gesture recognition has attracted increasing attention because skeletal representations provide compact structural cues while being less sensitive to background appearance and illumination variations than RGB-based methods [1], [2]. In addition, skeleton representations are computationally efficient and suitable for real-time human-computer interaction applications.

However, hand gestures involve subtle finger articulations and small spatial differences among joints, making skeleton-based recognition highly sensitive to viewpoint variations and self-occlusion [4], [6]. Under hand-in-the-wild conditions, such variations can distort skeleton coordinate distributions and degrade cross-dataset generalization performance [4]–[6]. Therefore, viewpoint robustness remains a fundamental challenge for skeleton-based hand gesture recognition.

B. Viewpoint robustness and cross-view learning

Learning viewpoint-invariant representations remains an important problem in skeleton-based action and gesture recognition, as the same gesture can produce substantially different coordinate distributions across camera viewpoints [4], [9], [10]. Cross-view and multiview methods have therefore been widely investigated to improve viewpoint robustness. Classical approaches such as canonical correlation analysis learn shared latent spaces by maximizing correlations among views [3], while multiview discriminant analysis introduces class-discriminative constraints for cross-view alignment [4]. More recent studies employ viewpoint-guided neural architectures and contrastive learning to learn viewpoint-invariant skeleton representations [9], [10].

However, practical hand gesture recognition systems usually provide only a single observed viewpoint during testing, making methods that depend on multiview observations at inference less applicable. This motivates the use of synthetic viewpoints to increase viewpoint diversity during training while preserving single-view inference.

C. Synthetic viewpoint generation for representation learning

Synthetic data generation has been widely used to improve viewpoint generalization when real multiview data collection is expensive or impractical. In action recognition, synthetic human rendering improves recognition from unseen viewpoints using systematically generated camera views [5], [12]. Unlike simple geometric rotation, virtual camera-based viewpoint generation explicitly models viewpoint-dependent camera-space transformations through a controlled virtual-camera system. Such synthetic multiview supervision has

recently been explored for viewpoint invariant skeleton representation learning [9].

For hand pose estimation and hand gesture recognition, constructing large-scale annotated 3D datasets remains challenging due to annotation complexity, occlusion, and dataset bias [6], [11]. Motivated by these challenges, this work employs a Blender-based virtual camera system to generate camera-space skeleton views from original hand skeletons for training-time viewpoint supervision.

D. Contrastive and consistency-based representation learning

Consistency-based and contrastive learning have been widely explored for invariant representation learning under different data transformations and viewpoints. In supervised settings, label information can be used to define positive and negative pairs, leading to supervised contrastive learning [7]. For skeleton-based action and hand gesture recognition, contrastive objectives have also been investigated to improve viewpoint-invariant representation learning across different views [6], [8], [9].

In this work, supervised contrastive learning is evaluated as an auxiliary regularization strategy. However, the proposed framework mainly focuses on original-anchored multiview consistency learning, where synthetic camera-space embeddings are aligned with the original skeleton embedding to reduce viewpoint-induced representation variations while preserving practical single-view inference.

III. PROPOSED METHOD

This section presents the proposed original-anchored multiview consistency framework for 3D skeleton-based hand gesture recognition, as illustrated in Fig. 2. Synthetic camera-space skeleton views are used as training-time supervision while preserving single-view inference during testing.

A. Hand gesture definition and dataset collection

In this study, seven well-defined static hand gestures are considered: *ok*, *paper*, *rock*, *scissors*, *the-finger*, *thumbup*, and *thumbedown*, as illustrated in Fig. 3. To evaluate gesture recognition performance, two datasets containing these predefined gestures were collected, namely *CanonicalSet* and *HandinWildSet*. *CanonicalSet* was collected under controlled laboratory conditions, where participants followed standardized gesture instructions. In contrast, *HandinWildSet* was collected in natural scenes under less constrained conditions, resulting in greater viewpoint, orientation, and domain variations. Therefore, *HandinWildSet* is used exclusively for external cross-dataset evaluation.

CanonicalSet contains 6,547 samples, including 4,582 training, 982 validation, and 983 internal test samples, while *HandinWildSet* contains 675 external test samples. Across the seven classes, the sample counts range from 325 to 1,413 in *CanonicalSet* and from 52 to 123 in *HandinWildSet*. Reliable subject identifiers are unavailable in the current metadata; therefore, subject-level statistics and subject-independent splits are not reported. As illustrated in Fig. 4, *HandinWildSet*

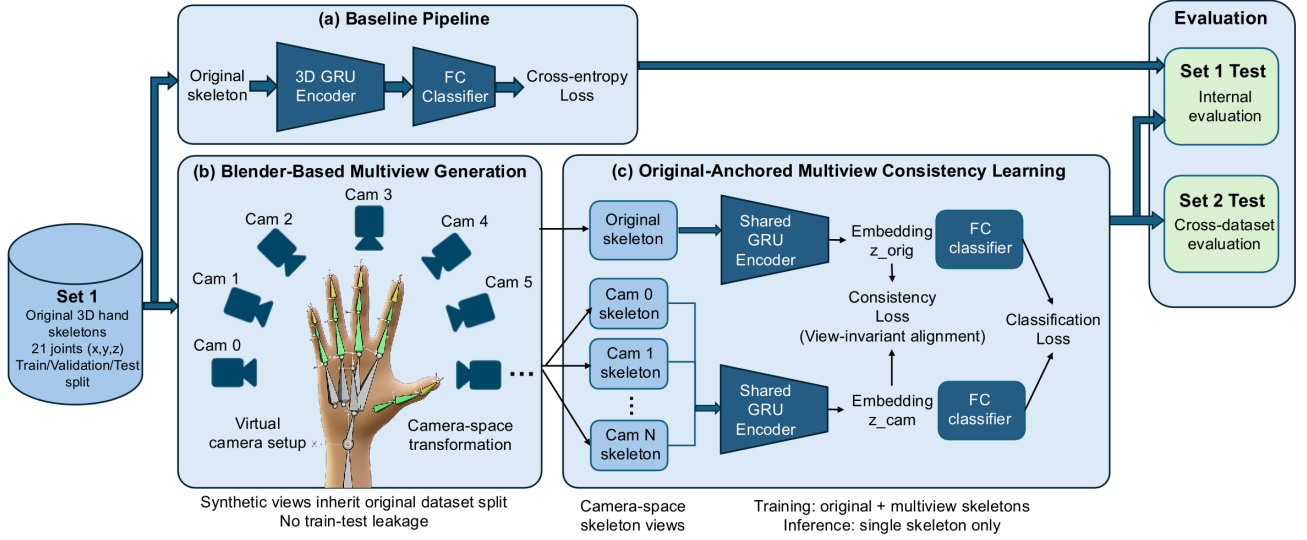


Fig. 2: Overall research framework from Blender-based hand animation and synthetic multiview generation to viewpoint-robust skeleton representation learning and cross-dataset evaluation.

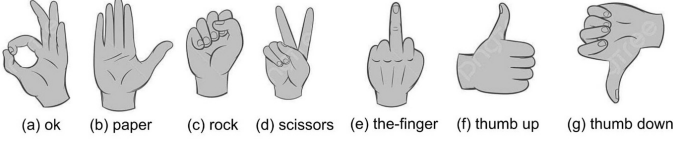


Fig. 3: Schematic views of seven hand gestures defined in this study.

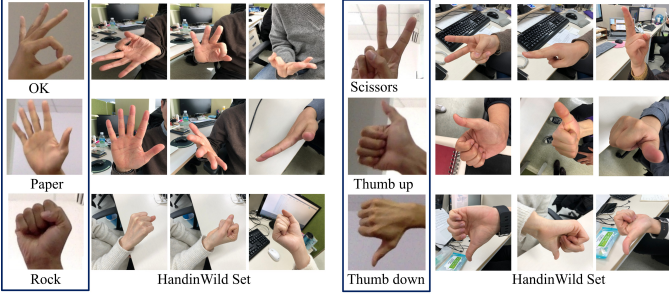


Fig. 4: Illustrations of 6/7 gestures in the HandinWildSet dataset with large viewpoint and appearance variations. Blue-framed images show the corresponding canonical references.

exhibits substantially larger variations in viewpoint, hand orientation, background, illumination, and finger visibility than *CanonicalSet*. For both datasets, MediaPipe Hands [14] is used to extract 21 hand landmarks from each RGB image, with each landmark represented by a coordinate triplet (x, y, z) .

B. Problem formulation

This work investigates cross-dataset hand gesture recognition under domain and viewpoint variations using two datasets with distinct visual characteristics. Each gesture is represented as a 3D hand skeleton consisting of $J = 21$ joints, where each joint is defined by its spatial coordinates (x, y, z) . A skeleton

sample is denoted as $\mathbf{s} \in \mathbb{R}^{J \times 3}$. Given an input skeleton $\mathbf{s}$, the objective is to predict its corresponding gesture label $c \in \{1, 2, \dots, C\}$, where $C = 7$ represents the total number of predefined gesture classes.

To improve viewpoint robustness, a Blender-based multiview generation pipeline is introduced to synthesize camera-space skeleton views during training [5]. For each original skeleton sample $\mathbf{s}^o$, the pipeline generates a set of synthetic camera-space observations as defined in Eq. (1):

$$\mathcal{V}(\mathbf{s}^o) = \{\mathbf{s}^1, \mathbf{s}^2, \dots, \mathbf{s}^N\} \quad (1)$$

where $\mathbf{s}^i$ denotes the skeleton observed from the i -th virtual camera viewpoint, and N is the total number of virtual views. Both 4-camera and 8-camera configurations are investigated, where the 8-camera setting introduces additional front-side viewpoints to further increase viewpoint diversity.

C. Blender-based camera-space skeleton generation

To increase viewpoint diversity during training, we employ a Blender-based hand animation and virtual-camera framework to generate synthetic camera-space skeletons. Each original 3D skeleton is first spatially normalized and mapped onto a rigged hand model to reconstruct the corresponding pose. Multiple virtual cameras are then placed around the reconstructed hand to capture different views of the same gesture instance. For each virtual camera i , the original skeleton $\mathbf{s}^o$ is transformed from the canonical coordinate system into the corresponding camera coordinate system as presented in Eq. (2):

$$\mathbf{s}^i = T_i(\mathbf{s}^o) \quad (2)$$

where $T_i(\cdot)$ denotes the geometric transformation associated with the i -th virtual camera. The resulting skeleton $\mathbf{s}^i$ preserves the gesture label while introducing viewpoint-dependent spatial variations. As shown in Fig. 5(a), a front-biased camera

configuration is adopted instead of a full 360-degree arrangement to increase viewpoint diversity while preserving gesture visibility. Fig. 5(b) presents representative rendered examples of the same gesture observed from different virtual viewpoints.

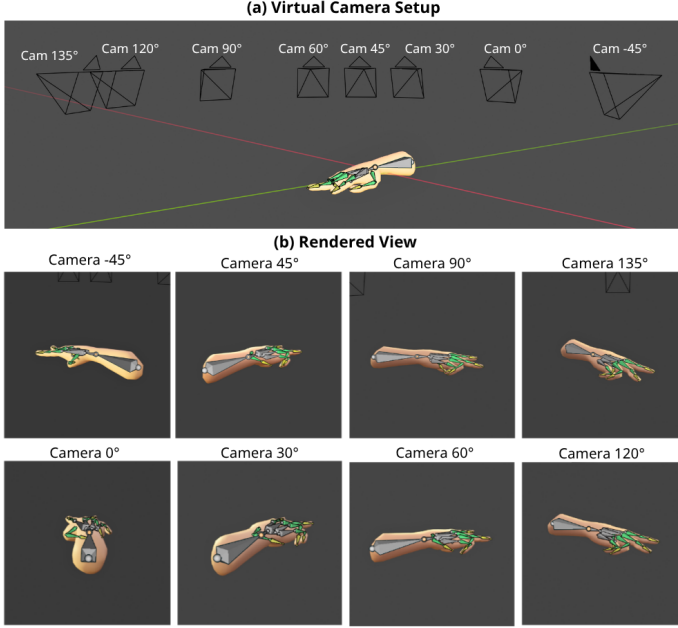


Fig. 5: Virtual camera configuration for synthetic multiview skeleton generation.

D. GRU-based skeleton representation learning

1) *Single-view skeleton encoder*: The basic hand gesture recognition model is a single-view GRU-based skeleton encoder which is illustrated in Fig. 2(a). A hand skeleton is treated as an ordered sequence of 21 joints, each joint is denoted by $\mathbf{p}_j = (x_j, y_j, z_j)$ and corresponds to the 3D spatial coordinates of the j -th hand joint, as defined in Eq. (3):

$$\mathbf{P} = \{\mathbf{p}_j\}_{j=1}^{21}, \quad \mathbf{p}_j = (x_j, y_j, z_j) \in \mathbb{R}^3 \quad (3)$$

In this static gesture setting, the sequence dimension corresponds to the predefined ordering of hand joints rather than consecutive temporal frames. The ordered joint sequence is processed by a GRU encoder [13] as shown in Eq. (4):

$$H = \{\mathbf{h}_j \in \mathbb{R}^{128}\}_{j=1}^{21} = \mathcal{M}_{GRU} \left(\{\mathbf{p}_j \in \mathbb{R}^3\}_{j=1}^{21} \right) \quad (4)$$

where $\mathbf{h}_j$ denotes the hidden state at joint position j , $\mathbf{h}_j = \mathcal{M}_{GRU}(\mathbf{p}_j, \mathbf{h}_{j-1})$, ($j = 1, 2, \dots, 21$).

The global skeleton representation is obtained from the final hidden state after the GRU has processed the complete 21-joint sequence, as shown in Eq. (5):

$$\mathbf{z} = \mathbf{h}_{21}, \quad \mathbf{z}, \mathbf{h}_{21} \in \mathbb{R}^{128} \quad (5)$$

The feature vector $\mathbf{z} \in \mathbb{R}^{128}$ is then passed through a fully connected classification layer that maps the 128-dimensional embedding to C gesture classes, and the network is optimized using the standard Cross-Entropy (CE) loss.

2) *Original-anchored multiview consistency learning*: Direct multiview learning typically requires multiple synchronized views during both training and inference. However, this setting is impractical for cross-dataset evaluation because external datasets usually provide only a single observed skeleton view. To address this limitation, an original-anchored multiview consistency learning framework is proposed, which leverages synthetic multiview supervision during training while preserving single-view inference capability as illustrated in Fig. 2(b) and (c).

Given an original skeleton sample $\mathbf{s}^o$ and its generated camera-space views $\{\mathbf{s}^1, \mathbf{s}^2, \dots, \mathbf{s}^N\}$ (Fig. 2(b)), all skeletons are processed independently using a shared GRU encoder as shown in Eq. (6) and Eq. (7), respectively:

$$\mathbf{z}^o = \mathcal{M}_{GRU}^\theta(\mathbf{s}^o) \quad (6)$$

$$\mathbf{z}^i = \mathcal{M}_{GRU}^\theta(\mathbf{s}^i), \quad i = 1, \dots, N \quad (7)$$

where $\mathcal{M}_{GRU}^\theta$ denotes the GRU encoder with shared parameters θ . The latent feature representations of the original skeleton and the generated camera-space views are denoted by $\mathbf{z}^o$ and $\mathbf{z}^i$ (Fig. 2(c)).

3) *Optimization objectives*: The learned feature representations are subsequently fed into a shared fully connected classification layer. The classification objective is computed over both the original skeleton and all generated views using cross-entropy loss as computed in Eq. (8):

$$\mathcal{L}_{cls} = \frac{1}{N+1} \left[CE(\mathbf{o}^o, y) + \sum_{i=1}^N CE(\mathbf{o}^i, y) \right] \quad (8)$$

When only the classification objective $\mathcal{L}_{cls}$ is used, the model is denoted as **MV-CE**. It uses both the original skeleton and synthetic camera-space views during training, without consistency or contrastive regularization, isolating the effect of synthetic multiview augmentation.

In this paper, two regularization strategies are investigated as follows:

Consistency regularization strategy: To encourage viewpoint-invariant representation learning, the camera-space embeddings are aligned with the original skeleton embedding through consistency regularization. The original skeleton is used as the anchor because it corresponds to the representation used in single-view inference. The consistency loss is formulated using cosine distance as shown in Eq. (9):

$$\mathcal{L}_{con} = \frac{1}{N} \sum_{i=1}^N \left(1 - \frac{\mathbf{z}^o \cdot \mathbf{z}^i}{\|\mathbf{z}^o\|_2 \|\mathbf{z}^i\|_2} \right) \quad (9)$$

The first optimization objective is defined as Eq. (10):

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda \mathcal{L}_{con} \quad (10)$$

where λ controls the consistency contribution. Aligning each camera-space embedding with the original-view representation encourages viewpoint-invariant learning while preserving single-view inference.

Supervised contrastive strategy: Furthermore, a supervised contrastive learning (SupCon) variant is additionally investigated as an alternative representation regularization strategy. Specifically, the learned embeddings are optimized using a combination of cross-entropy loss and supervised contrastive loss [7] as presented in Eq. (11):

$$\mathcal{L}_{sup} = \sum_{k=1}^B \frac{-1}{|P(k)|} \sum_{p \in P(k)} \log \frac{\exp(\text{sim}(\mathbf{z}_k, \mathbf{z}_p)/\tau)}{\sum_{a \in A(k)} \exp(\text{sim}(\mathbf{z}_k, \mathbf{z}_a)/\tau)} \quad (11)$$

where $P(k)$ denotes the set of positive samples that share the same label as sample k , $A(k)$ denotes all samples except k , $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, and τ is the temperature parameter. The second optimization objective is computed as Eq. (12):

$$\mathcal{L} = \mathcal{L}_{cls} + \alpha \mathcal{L}_{sup} \quad (12)$$

where α controls the supervised contrastive contribution. MV-SupCon is evaluated as an alternative representation regularization strategy and compared with original-anchored consistency in Section IV.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Protocol and evaluation setup

The *CanonicalSet* is partitioned into training, validation, and test subsets using the original skeleton samples. Blender-based multiview generation is performed independently within each subset, so that every synthetic camera-space skeleton remains in the same partition as its corresponding original sample, thereby preventing cross-view train-test leakage. *HandinWildSet* is excluded from both training and multiview generation and is used solely for external cross-dataset evaluation. For each training run, model checkpoints are selected exclusively according to validation accuracy on *CanonicalSet*.

For Blender-based synthetic multiview generation, two virtual camera configurations are investigated, namely 4-camera and 8-camera setups. The 4-camera setting uses azimuth angles of 0° , 45° , 90° , and 135° , while the 8-camera setting additionally includes -45° , 30° , 60° , and 120° . These front-side viewpoints increase viewpoint diversity while preserving gesture visibility.

All experiments are implemented in PyTorch 2.5.1 using the Apple MPS backend and random seed 42. The models are optimized with Adam (10^{-3}), a batch size of 64, and early stopping based on validation accuracy. The evaluated methods include the **3D GRU baseline**, **ROT3D**, **MV-CE**, the proposed **MV-Consistency Anchor**, and **MV-SupCon Anchor**. All multiview variants use the same GRU backbone to provide a controlled comparison of training strategies.

B. Effect of regularization weights λ and α

The effects of the consistency weight λ and supervised contrastive weight α are evaluated on *CanonicalSet* and *HandinWildSet*, as summarized in Tab. I. For MV-Consistency, increasing λ from 0.05 to 0.3 improves *HandinWildSet* accuracy

TABLE I: Effect of consistency and supervised contrastive loss weights on recognition accuracy.

MV-Consistency			MV-SupCon		
λ	CanonicalSet Acc. (%)	HandinWildSet Acc. (%)	α	CanonicalSet Acc. (%)	HandinWildSet Acc. (%)
0.05	93.39	71.70	0.05	93.79	70.52
0.10	94.20	73.78	0.10	95.32	72.29
0.30	94.61	75.85	0.30	93.59	70.67
0.50	94.00	75.70	0.50	94.81	66.81
0.70	92.98	75.11	0.70	94.60	66.96
0.90	93.89	75.70	0.90	95.42	64.44
1.00	93.18	75.25	1.00	94.20	69.77

TABLE II: Experimental comparison under the 7-class protocol.

Method	Objective	CanonicalSet Acc. (%)	HandinWildSet Acc. (%)
3D GRU Baseline	CE	94.20	62.96
ROT3D	CE	95.93	70.22
MV-CE (4 views)	CE	95.22	59.56
MV-Consistency (4 views)	CE + Cons.	91.56	67.70
MV-SupCon (4 views)	CE + SupCon	93.90	64.44
MV-CE (8 views)	CE	94.51	68.44
MV-Consistency (8 views)	CE + Cons.	94.61	75.85
MV-SupCon (8 views)	CE + SupCon	95.32	72.29

from 71.70% to 75.85%. Although performance remains relatively stable between 75.11% and 75.70% for larger weights, none outperforms $\lambda = 0.3$. This suggests that a moderate consistency weight provides effective viewpoint alignment without overemphasizing the regularization objective.

For MV-SupCon, $\alpha = 0.1$ yields the highest *HandinWildSet* accuracy of 72.29%, together with 95.32% on *CanonicalSet*. Larger values generally reduce external accuracy while maintaining high source-domain performance, indicating that excessive class-level contrastive regularization may favor source-domain discrimination over cross-dataset generalization. Overall, MV-Consistency provides higher and more stable external performance than MV-SupCon. Therefore, $\lambda = 0.3$ and $\alpha = 0.1$ are selected for the reported 8-view configurations.

C. Overall results

Tab. II presents the overall comparison under the 7-class protocol. The 3D GRU baseline achieves high accuracy on *CanonicalSet* but drops substantially on *HandinWildSet*, from 94.2% to 62.96%, indicating limited cross-dataset generalization under single-view training. By introducing synthetic multiview supervision, the 8-view MV-CE model achieves 68.44% on *HandinWildSet*, while the proposed 8-view MV-Consistency Anchor further improves the accuracy to 75.85%, outperforming both the GRU baseline and ROT3D of 70.22%. This suggests that combining Blender-generated camera-space skeletons with original-anchored consistency provides more effective viewpoint supervision than simple geometric rotation.

Compared with MV-SupCon Anchor, MV-Consistency Anchor achieves better *HandinWildSet* accuracy in the 8-view setting (75.85% versus 72.29%), suggesting that original-anchored consistency is more effective in the evaluated cross-

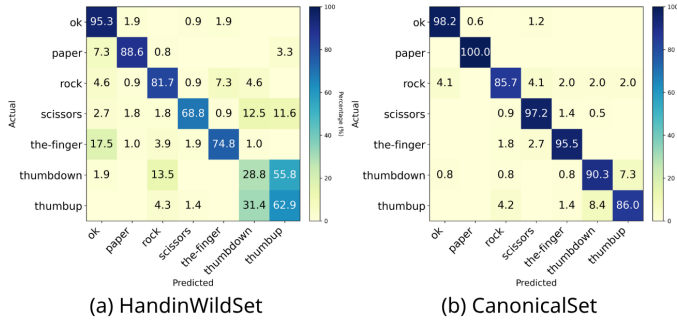


Fig. 6: Confusion matrices of the best-performing MV-Consistency anchor model under the 7-class protocol.

dataset setting. Fig. 6 presents the normalized confusion matrices of the best-performing MV-Consistency Anchor model. Recognition remains stable across most classes on *CanonicalSet*, whereas errors on *HandinWildSet* are concentrated in viewpoint-sensitive gestures. In particular, *thumbup* and *thumbedown* are frequently confused due to their strong dependence on hand orientation, while errors involving *rock* and *the-finger* indicate the remaining challenges caused by partial finger visibility and oblique viewpoints.

Effect of the number of synthetic cameras: As shown in rows 3–8 of Tab. II, the effect of increasing the number of synthetic camera views is analyzed. For all three multiview learning strategies, the 8-camera setting outperforms the 4-camera setting on *HandinWildSet*, confirming that increasing viewpoint diversity during training improves cross-dataset robustness. The improvement is particularly high for MV-Consistency Anchor, where *HandinWildSet* accuracy increases from 67.7% to 75.85%. Compared with the single-view 3D GRU baseline, the final 8-camera consistency model improves *HandinWildSet* accuracy from 62.96% to 75.85%. These results show that the proposed framework benefits from two factors simultaneously: synthetic multiview generation and original-anchored consistency alignment.

Preliminary human evaluation: An initial human evaluation was conducted on *HandinWildSet* using real RGB images with 10 evaluators performing manual gesture labeling. Each evaluator labeled an assigned subset of images, resulting in 3,840 image-level assignments. The mean accuracy across evaluators was 95.98%, with a standard deviation of 4.16 percentage points, which remains substantially higher than the best model accuracy of 75.85%. The remaining gap suggests that human observers may exploit additional contextual information, including palm orientation, global hand appearance, and scene-level cues, particularly for ambiguous gestures such as *thumbup* and *thumbedown*.

V. CONCLUSION

This paper presents an original-anchored multi-view consistency framework for hand-in-the-wild gesture recognition. The proposed method utilizes Blender-generated camera-space skeleton views as training-time supervision to enhance viewpoint robustness. *CanonicalSet* is used for training and syn-

thetic multi-view generation, whereas *HandinWildSet* is used for external cross-dataset evaluation. Experimental results under the 7-class protocol demonstrate that synthetic multi-view supervision substantially improves viewpoint robustness and cross-dataset generalization. The proposed 8-camera consistency model achieves the best accuracy of 75.85% on *HandinWildSet*, outperforming single-view training, the clean ROT3D baseline, and supervised contrastive learning variants. The corresponding 8-camera model without consistency achieves 68.44%, further supporting the benefit of original-anchored consistency. As a preliminary analysis, human evaluation with 10 evaluators achieves a mean accuracy of 95.98% with a standard deviation of 4.16 percentage points, revealing that a noticeable performance gap still exists between human recognition capability and the proposed method. These findings suggest that incorporating synthetic contextual information may represent a promising direction for improving viewpoint-robust hand-in-the-wild gesture recognition.

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